

Flag 6 (Login 3)

Analysis & Exploitation Steps:-

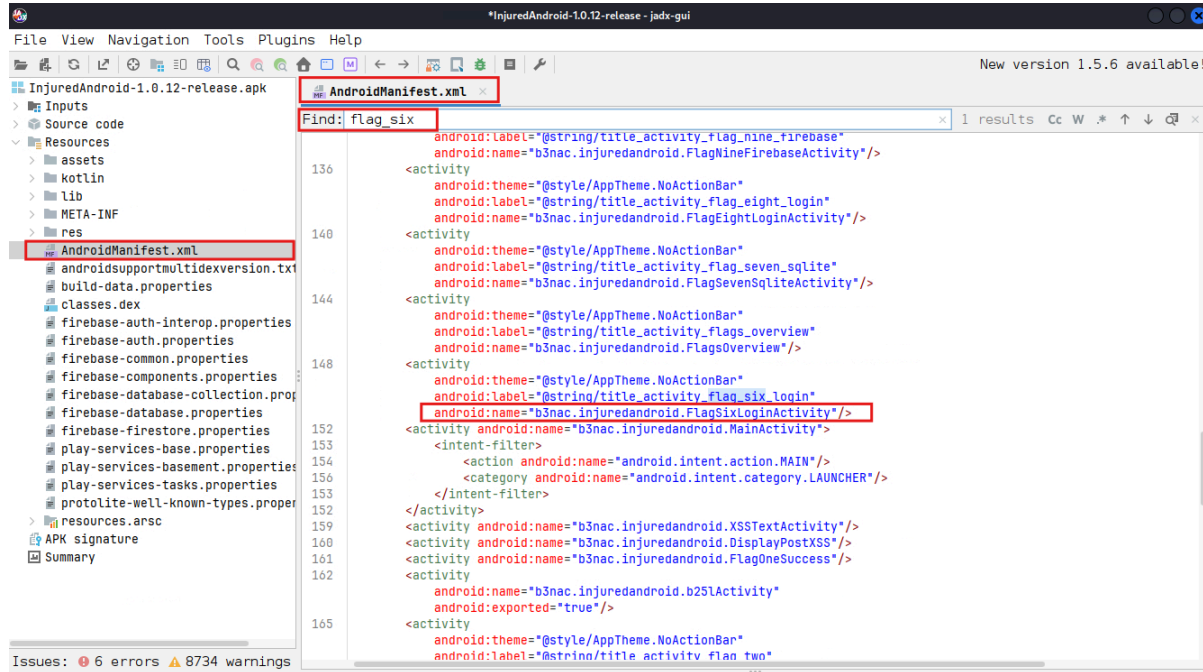


Figure 1: Analyzing AndroidManifest.xml Searching for the String "flag_six" and Found an Activity of Class "b3nac.injuredandroid.FlagSixLoginActivity"

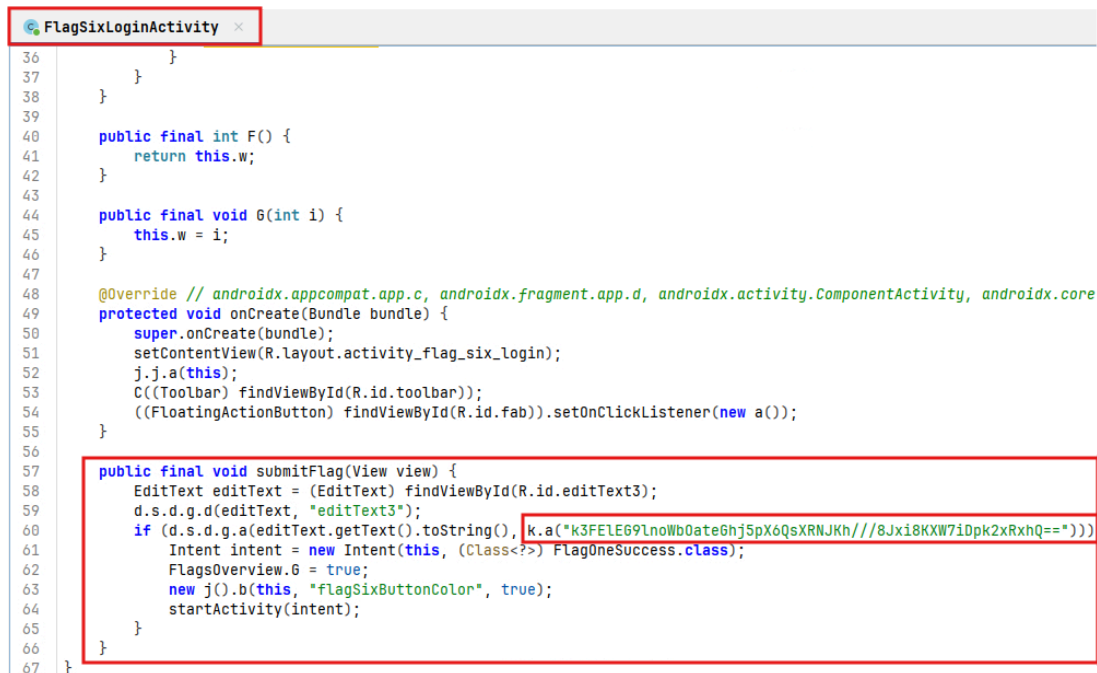


Figure 2: Displays the Decompiled Source for "b3nac.injuredandroid.FlagSixLoginActivity" Class

```

1 package b3nac.injuredandroid;
2
3 import android.util.Base64;
4
5 /* JADX INFO: loaded from: classes.dex */
6 public class h {
7
8     /* JADX INFO: renamed from: a, reason: collision with root package name */
9     private static byte[] f1469a = Base64.decode("Q2FwdHVyM1RoMXM=" 0);
10
11     /* JADX INFO: renamed from: b, reason: collision with root package name */
12     private static byte[] f1470b = Base64.decode("e0NhCHR1cjNUaDFzV69vfQ==", 0);
13
14     /* JADX INFO: renamed from: c, reason: collision with root package name */
15     private static String f1471c = "9EEADi^?:?FC652?5C@:5]7:C632D6:@]4@>^DB=:E6];D@?";
16
17     static byte[] a() {
18         return f1470b;
19     }
20
21     static byte[] b() {
22         return f1469a;
23     }
24
25     static String c() {
26         return f1471c;
27     }
28 }

```

Figure 3: Displays Encoded Encryption/Decryption Key

```

$ python3
Python 3.13.12 (main, Feb  4 2026, 15:06:39) [GCC 15.2.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
>>> import base64
...
... from Crypto.Cipher import DES
...
... ENCODED_KEY = "Q2FwdHVyM1RoMXM="
... ENCODED_STRING = "k3FELEG9lnowB0ateGhj5pX6QsXRNJKh///8Jxi8KXW7iDpk2xRxhQ=="
...
... def decrypt():
...     key_byte_array = base64.b64decode(ENCODED_KEY)[:8]
...     string_byte_array = base64.b64decode(ENCODED_STRING)
...     des_handler = DES.new(key_byte_array, DES.MODE_ECB)
...
...     print(des_handler.decrypt(string_byte_array))
...
... if __name__ == '__main__':
...     decrypt()
...
b"{This_Isn't_Where_I_Parked_My_Car}\x06\x06\x06\x06\x06\x06"
>>> |

```

Figure 4: Displays Successful Decryption Using the Encoded Decryption Key

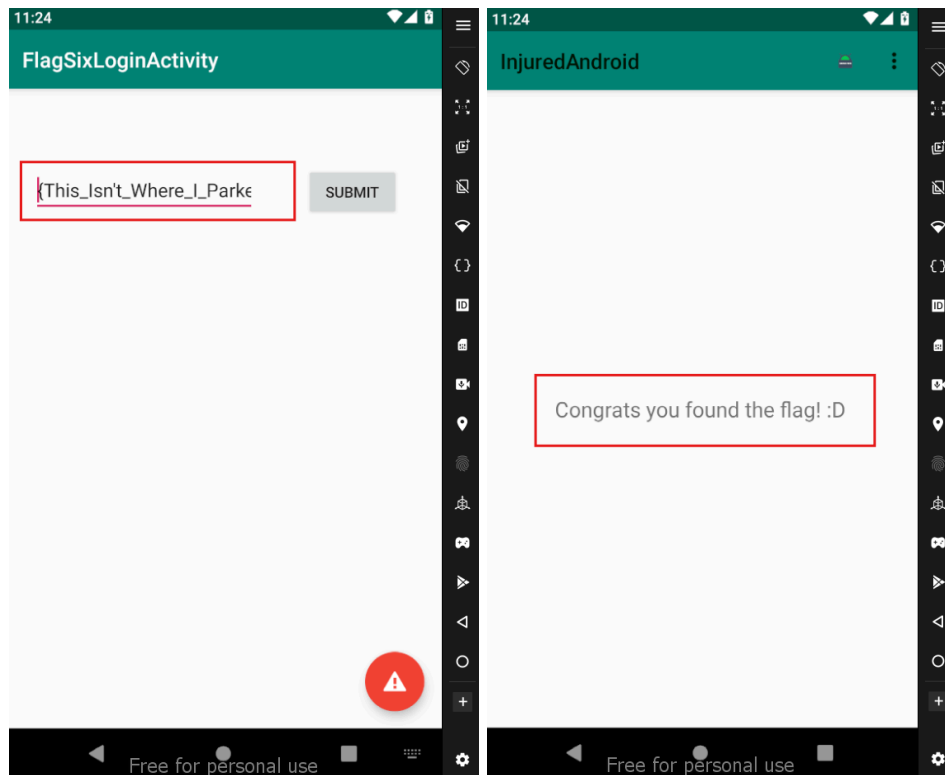


Figure 5: Displays Successful Submission of the Flag

Impacts:-

- Hardcoded encryption keys can be extracted through static analysis & also attackers can use dynamic instrumentation to intercept decrypted sensitive data

Remediations:-

- Never hardcode encryption keys in the APK, store keys securely and use Android Keystore where appropriate & protect sensitive data and minimize exposure during runtime.